
Network Swiss Army Knife (NSAK)

Bachelor Thesis · BFH-TI 2026

June 9, 2026

Lukas von Allmen (vonall3) and Frank Gauss (gaussf1) | Bern University of Applied Sciences

The Challenge

Rising Cyber Threats

- Attackers already leverage AI — defenders must keep pace
- Red & blue team exercises demand deep expertise
- High recurring effort — inaccessible for small and medium-sized enterprises
- Scripted automation is brittle and inflexible

Hypothesis

Integrating **agentic AI** into NSAK improves **adaptability, flexibility** and **automation** over static scenarios, while **lowering the operator's required expertise** to interpret results and plan follow-up steps.



Research Questions

- Can an AI agent match a scripted scenario at automated network reconnaissance?
- How effectively can AI assist a human operator during a live security investigation?
- Does it hold up in real-world networks beyond a controlled lab?

Our Approach



Agentic AI

- › Building upon the predecessor project: Network Swiss Army Knife
NSAK — A modular framework for network security scenario automation
- › Integrate an LLM agent into NSAK with LangChain
- › Equip it with tool calling for autonomous execution
- › Benchmark against a scripted reference scenario

3 LLMs • 2 Network Environments • Static Reference Baseline

What We Built



AI Reconnaissance Scenario



Static Reference Scenario



Reproducible Network Environment



Benchmark Suite



And more...

Evaluation Highlights



Containerlab (Virtual)

- ✓ All 3 models completed reconnaissance autonomously
- ✓ Multi-agent decomposition improved smaller models significantly
- Larger models showed no benefit from decomposition



BFH Network Security Lab (Physical)

- ✓ Frontier model navigated a live network with 35+ hosts
- Structured output failed — adapted to unstructured mode
- Small local model (qwen3:8b) could not complete the task

Static scenario: fast & reproducible, but no reasoning or report generation

Demo: Interactive AI Agent



Approval workflow

- › Operator confirms each drill before execution



AI reasoning

- › Model selects tools, adapts strategy, explains decisions



Email report

- › Prioritized findings sent as a human-readable summary

Demo: Interactive AI Agent

Prompt:

Help met to find all vulnerable services on interface eth1

Demo: Interactive AI Agent

Prompt:

Please explain what the leaked documents are revealing

Demo: Interactive AI Agent

Prompt:

I think we gathered enough information now, create a summary in Markdown and send it as an email

Key Findings



AI Scenario

- ✓ Prioritized, natural-language reports
- ✓ Adaptive to unknown environments
- ✓ Low barrier — minimal expertise needed
- Variable output, hallucination risk
- Requires human supervision



Static Scenario

- ✓ Scripted & reproducible
- ✓ Fast, no token cost
- ✓ Precise, auditable steps
- No reasoning or report generation
- Requires domain expertise to build

“AI and scripted automation are complementary — not competing — approaches.”